



BEYOND THE CLASSROOM - WHAT DRIVES STUDENT DROPOUT?

Identifying key factors and predicting student dropout risk

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01 | INTRODUCTION

Student dropout can be influenced by academic performance, engagement, well-being, socioeconomic conditions and accessibility. Identifying factors associated with dropout can help institutions recognise students who may require additional support.

RESEARCH QUESTION

Which student characteristics are associated with dropout, and can they be used to predict dropout risk?

02 | DATASET



10,000 students

19 variables

1500 sample

SAMPLING - STRATIFIED BY DROPOUT STATUS

- 9500 Complete Cases
- Proportional Stratification
- Random Sampling
- 1500 Students

Target - DROPOUT

Dropout: 0 = Retained | 1 = Dropout

Source: Kaggle -

[Student Dropout Prediction - Dataset](#)

03 | METHODOLOGY



- Data
- Quality Check
- Stratified Sampling
- Descriptive Analysis
- Statistical Inference
- Logistic Regression
- Model Evaluation

08 | CONCLUSIONS

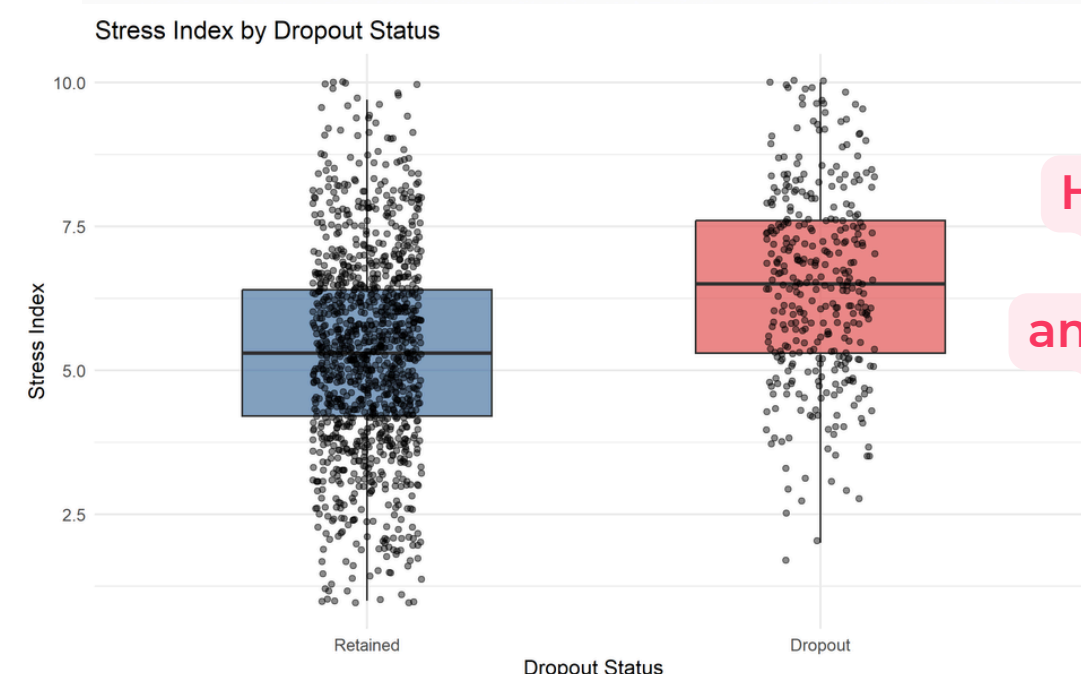
Academic performance, attendance, stress and assignment delays were associated with dropout.

Early identification can help target academic and wellbeing support.

Monitor attendance, academic performance, stress and assignment delays regularly and provide support for students with high stress.



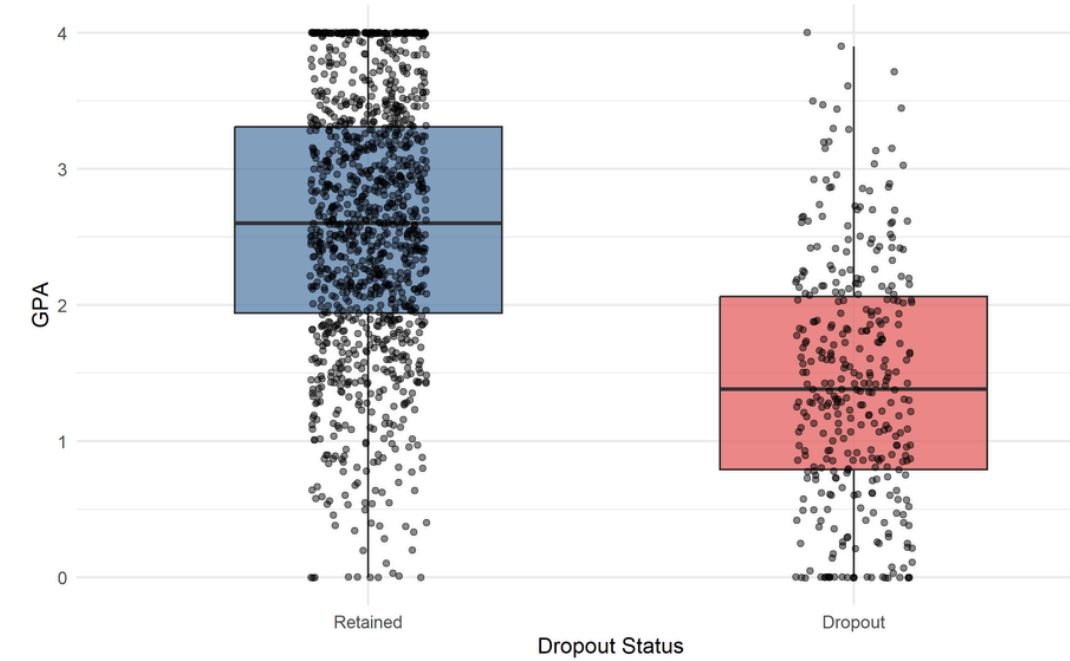
Variable	Mean	Median	Standard Deviation	Interquartile Range
GPA	2.31	2.37	1.04	1.55
Attendance Rate (%)	81.82	81.70	8.02	10.73
Study Hours Per Day	4.05	4.00	1.28	1.73
Stress Index	5.59	5.60	1.80	2.40
Assignment Delay (Days)	1.83	2.00	1.37	2.00



Higher stress levels are observed among dropout students.

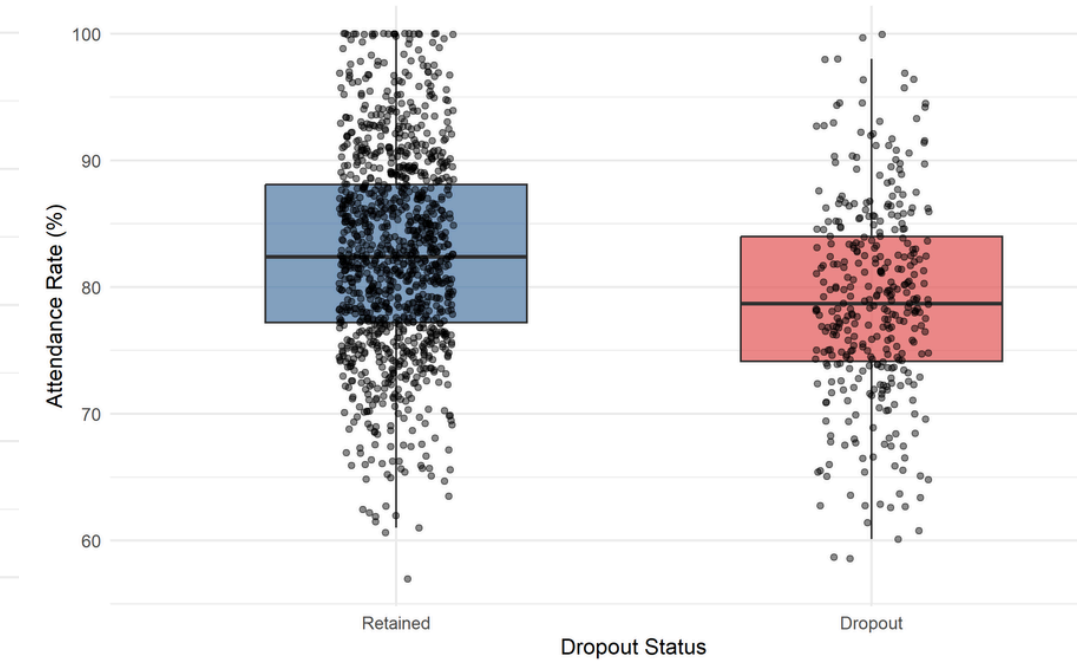
04 | DESCRIPTIVE ANALYSIS

GPA Distribution by Dropout Status



Dropout students show lower GPA values than retained students.

Attendance Rate by Dropout Status



Lower attendance is observed among dropout students.

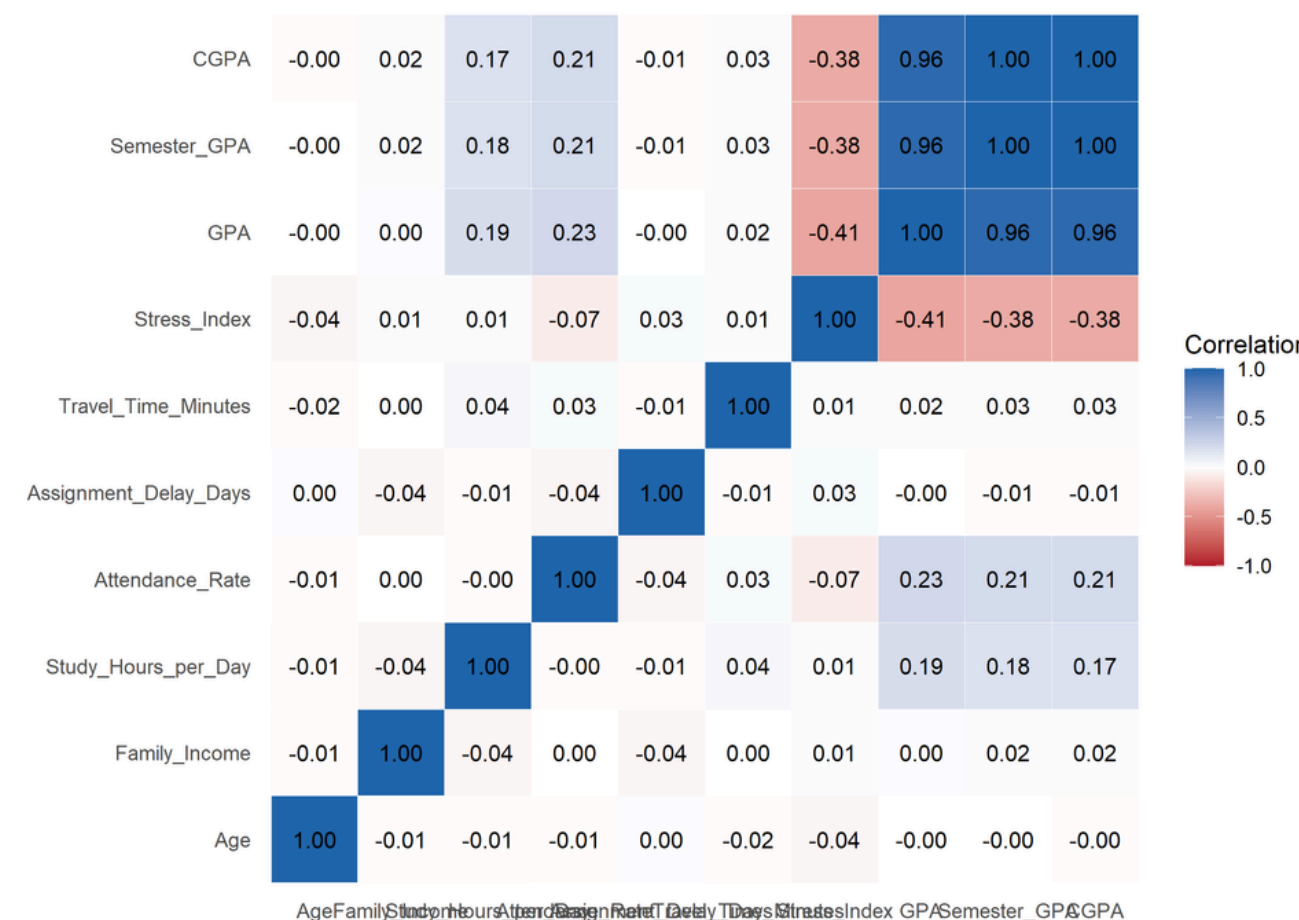
Attendance <-> GPA
 $r = 0.231$
WEAK POSITIVE CORRELATION

Academic performance measures (GPA, Semester GPA and CGPA) are strongly correlated because they represent closely related measures of academic performance.

Greater assignment delays were also observed among dropout students.

Correlation indicates association, not causation.

Correlation Heatmap of Numerical Variables



Hypothesis tests were conducted at $\alpha = 0.05$ to assess statistically significant differences and associations.

05 | STATISTICAL INFERENCE

Analysis	Test	Statistic	p-value	Decision
GPA × Dropout	Welch t-test	$t = 20.867$	$< 2.2 \times 10^{-16}$	Reject H_0
Attendance × Dropout	Welch t-test	$t = 7.783$	3.18×10^{-14}	Reject H_0
Stress × Dropout	Welch t-test	$t = -11.534$	$< 2.2 \times 10^{-16}$	Reject H_0
Study Hours × Dropout	Welch t-test	$t = 3.581$	0.00037	Reject H_0
Assignment Delay × Dropout	Welch t-test	$t = -3.474$	0.00055	Reject H_0
Internet × Dropout	Chi-square test	$\chi^2 = 3.014$	0.0825	Fail to reject H_0
Attendance × GPA	Pearson correlation	$r = 0.231$	$< 2.2 \times 10^{-16}$	Reject H_0
GPA × Year	One-way ANOVA	$F = 0.36$	0.782	Fail to reject H_0

95% CONFIDENCE INTERVAL

23.60%

Estimated Dropout Rate:
95% CI: 21.49% – 25.85%

07 | RESULTS

ACADEMIC & ENGAGEMENT

Dropout students showed lower GPA and attendance levels than retained students.

STUDENT FACTORS

Dropout students showed higher stress levels and greater assignment delays.

STATISTICAL EVIDENCE

GPA, attendance, stress, study hours and assignment delays differed significantly between groups, while Internet access and academic year showed no significant relationship with dropout.

PREDICTIVE MODEL

Logistic regression achieved 83.28% accuracy and a ROC-AUC of 0.8365.

09 | REFERENCES

- [1] SLIIT. (2026). IE2024 Probability & Statistics – Individual Assignment Brief.
- [2] Ali, M. (n.d.). Student Dropout Prediction Dataset. Kaggle
- [3] Rathnayake, S. C. (2026). IE2024 Probability and Statistics – Lab Sheet 05: Advanced Descriptive Analysis and Data Visualization with ggplot2. SLIIT.

06 | PREDICTIVE MODELLING

Logistic Regression



Target: Dropout
0 = Retained | 1 = Dropout



Prediction Threshold:
0.50



Data plot:
80% Train | 20% Test

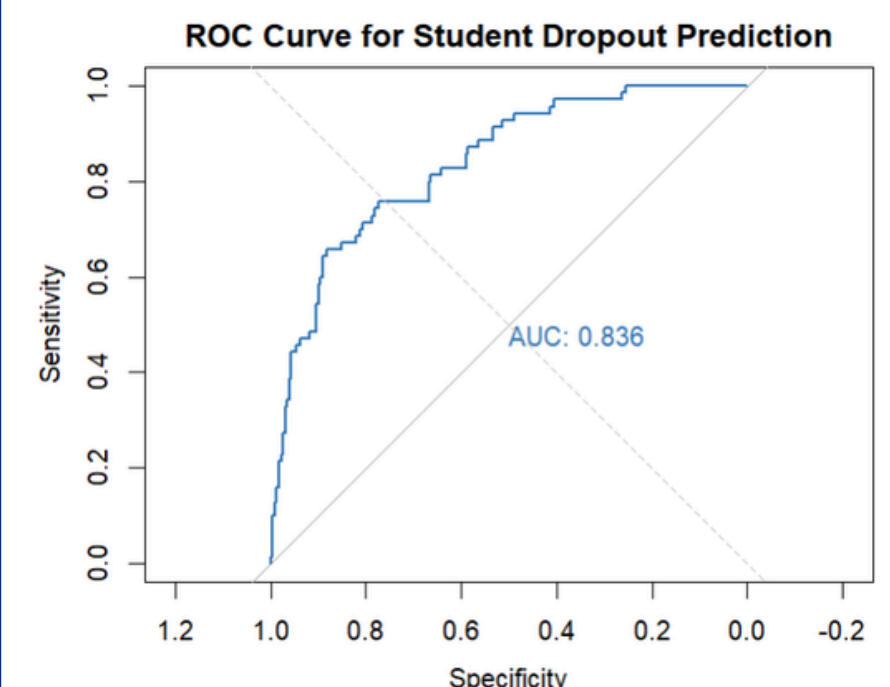
Key Predictors Associated with Dropout

Predictor	Association with dropout
GPA	Lower odds of dropout
Attendance	Lower odds of dropout
Stress	Higher odds of dropout
Assignment Delay	Higher odds of dropout
Age	Higher odds of dropout
Year 3	Higher odds of dropout

Lower odds of dropout Higher odds of dropout

Predictors shown are statistically significant in the final model ($p < 0.05$).

Model Performance



Model Performance (continued)

Metric	Result
Accuracy	83.28%
Sensitivity (Recall)	45.71%
Specificity	94.76%
Precision	72.73%
F1-score	56.14%
ROC-AUC	0.8365

+6.69

accuracy vs majority class baseline